

Moeez Ahmad

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EDUCATION

Virginia Tech

Bachelor of Science in Computer Science, GPA: 3.6/4.0

Blacksburg, VA

December 2025

Coursework: Object-Oriented Programming, Data Structures & Algorithms, Computer Organization, Data Analytics & Visualization, Software Engineering, Operating Systems, Artificial Intelligence, Databases

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: React.js, Node.js, Express.js, Next.js, LangGraph, LangChain, PyTorch

Databases: PostgreSQL, MySQL, MongoDB, Supabase

Tools & Platforms: AWS, Git, GitHub, Docker, Kubernetes, Redis

EXPERIENCE

Software Engineer

Stealth Startup

February 2026 – Present

Remote

- Built and maintained a backend API with **15+** production endpoints, implementing core business logic
- Implemented secure authentication and authorization for user access and protected API endpoints
- Optimized database queries, indexes, and schema structure, reducing query latency by **30%** and improving overall database performance by **20%**

Software Engineer

South Asian Business Council Of Virginia

May 2026 – July 2026

Herndon, VA

- Architected an AI admin platform using RAG and a **4-agent orchestration** in LangGraph, automating workflows
- Built a RAG pipeline with LangChain and Pinecone across 100+ internal documents, reducing search time by **70%**
- Developed agents for meeting summaries, email drafting, and social media content generation, saving **5+** hrs/week

Software Engineer

Virginia Tech's Visionarium Lab

August 2025 – December 2025

Blacksburg, VA

- Developed RESTful backend APIs using **Node.js** and **Express.js** for an aquaponics app, serving data to frontend
- Optimized PostgreSQL database schemas with normalization and indexing, reducing query response time by **15%**
- Implemented automated API tests using Jest, achieving **95%** endpoint coverage and improving backend reliability

Machine Learning Research Assistant

Virginia Tech Department of Computer Science

January 2025 – August 2025

Blacksburg, VA

- Improved indoor obstacle-detection reliability by **25%** in complex environments by fine-tuning a YOLOv12 object detection model in PyTorch, achieving a **12%** increase in precision and **21%** improvement in recall
- Built SmartGuide, a real-time computer vision pipeline achieving **60ms** inference latency on obstacle detection, leveraging depth estimation for spatial awareness in a navigation
- Collected and cleaned a 10,000-image indoor dataset covering 12 obstacle classes, improving model generalization and cutting false negatives by **15%** during real-world testing

Undergraduate Teaching Assistant

Virginia Tech Department of Computer Science

August 2025 – December 2025

Blacksburg, VA

- Supported **100+** students in learning database concepts such as ER modeling, relational design, and SQL querying
- Led discussion and debugging sessions to resolve SQL and database management issues
- Guided students through complex database design and implementation challenges during office hours

PROJECTS

onthefly 🌀 | *JavaScript, Node.js, Express.js, React, PostgreSQL, SQL, Git*

- Built a full-stack travel planning app enabling users to create trips, browse a destinations and organize itineraries
- Developed a RESTful Express API controllers/routers for trip management and browsing destinations
- Implemented GitHub OAuth with Passport.js and a normalized PostgreSQL schema to support relational trip data

FitFindr 🌀 | *Python, Gradio, Groq API*

- Built an AI agent that searches listings, matches outfits, and generates fit cards from natural language requests
- Designed a planning loop that selects tools based on prior results, passing state across calls without user re-entry
- Implemented per-tool error handling and fallback messaging for empty or invalid results for all three tools